## 9. Perform the Rotation of an image in a clockwise and counterclockwise direction.

**Aim:**

To perform image processing by rotating an image in clockwise and counterclockwise directions using the OpenCV library.

**Procedure:**

- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

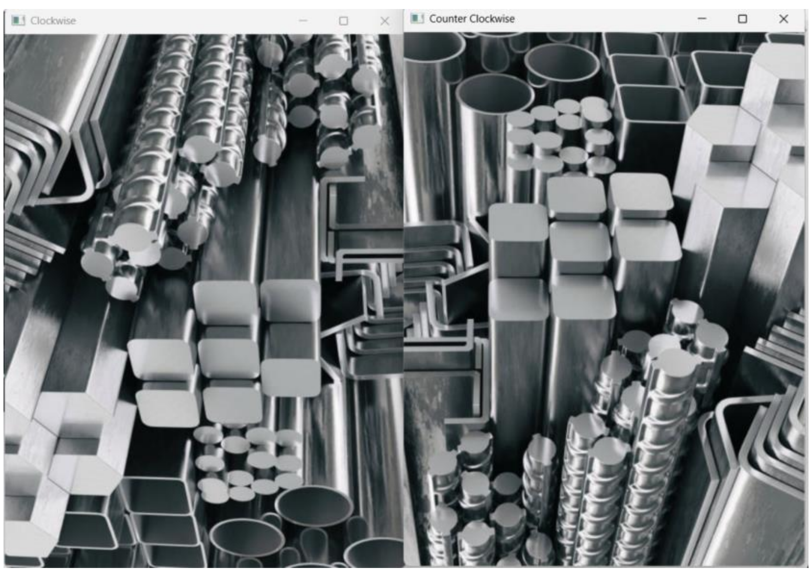
**Code:**

```
import cv2



# Read the image from Downloads folder

image =

cv2.imread("/Users/michellejoanna/Downloads/meee.jpeg")



# Check if the image is loaded successfully

if image is None:

    print("Image not found!")

    exit()



# Rotate 90° Clockwise
```

```
clockwise = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)


# Rotate 90° Counter Clockwise
counter_clockwise = cv2.rotate(image,
cv2.ROTATE_90_COUNTERCLOCKWISE)


# Display Images
cv2.imshow("Original Image", image)
cv2.imshow("90° Clockwise Rotation", clockwise)
cv2.imshow("90° Counter Clockwise Rotation",
counter_clockwise)


# Wait until a key is pressed
cv2.waitKey(0)


# Close all windows
cv2.destroyAllWindows()
```

**Sample Input**


**Sample Output**




**Result:**

The input image was successfully rotated in **clockwise and counterclockwise directions** using OpenCV.